SO-101 Training Framework
& Orchestrator Architecture

Phang Hong Hao (潘洪浩)

Tsinghua University

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Training Framework & Speed Benchmark

Two Policies on RTX 6000D

	PushT Diffusion
Policy	Diffusion (UNet)
Params	263M
Batch	64
Steps	100K

	SO-101 ACT
Policy	ACT (Transformer)
Params	84M
Batch	128 (max)
Steps	50K

torchcodec vs pyav

Config	Speed	Factor
PushT pyav	0.5–0.7 step/s	1x
PushT torchcodec	12–15 step/s	**20x**
ACT pyav 8w	0.38 step/s	1x
ACT pyav 8w AMP	0.45 step/s	1.2x
ACT torchcodec 8w AMP	**1.9 step/s**	**8.6x**

Key Finding

GPU 利用率始终 100%，**视频解码是真正瓶颈。** torchcodec GPU 硬件加速 » pyav CPU 逐帧解码。

Optimal Configuration & Known Bugs

Launch Command

Parameter	Value
video_backend	torchcodec
policy.type	act
policy.device	cuda (not cuda:N!)
chunk_size	100
dim_model	512
use_amp	True
batch_size	128 (max)
num_workers	8

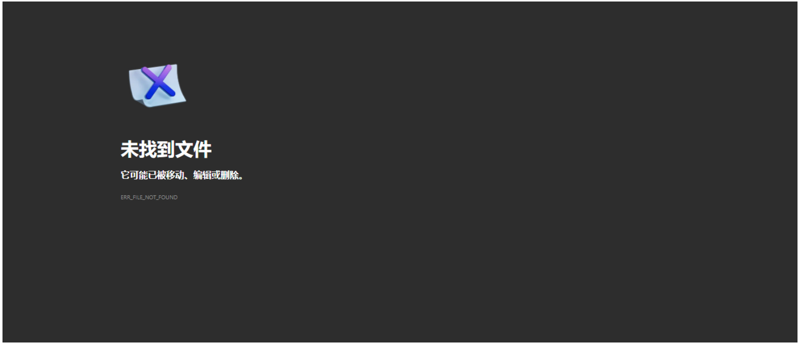
Why batch_size=128 Max?

ACT chunk_size=100 $O(n^2)$，128 batch 时显存 ~79GB（RTX 6000D 上限）。

Known Bugs & Workarounds

Bug	Fix
is_amp_available ValueError	CUDA_VISIBLE_DEVICES=6
torchcodec DecodingError	同上
CXXABI_1.3.15 not found	LD_PRELOAD=.../libstdc++.so.6
libavutil.so.X not found	conda install ffmpeg

Orchestrator Architecture



Core Modules

- `arm101_orchestrator.py` —主控 + CLI
- `phase_manager.py` —YAML 三层合并
- `state_store.py` —崩溃恢复 (原子 JSON)
- `data_collector.py` —MuJoCo 采集

Key Features

- `--fresh` / `--start-from <id>` / `--dry-run`
- 崩溃恢复: 重连训练 PID
- Loss Monitor: CONVERGED / OVERFITTING

Reference Models & Key Takeaways

HuggingFace Models (Survey 100+)

#	Model / Key Info
1	`Sa74ll/smolvla_so101` SmolVLA, 87.66% 成功率
2	`TakuyaHiraoka/act_so101` ACT, 真实机械臂
3	`davidlinjiahao/lerobot_so101` ACT, MuJoCo, 最相似

Pipeline

`scene.xml` → `MjModel` → trajectory → render → `LeRobotDataset` → HF Hub

Key Takeaways

1. **ACT 是 SO-101 主流策略**
   - HF 上 30+ ACT 模型
2. **SmolVLA 成功率更高**
   - Sa74ll 达 87.66%，靠数据分割
3. **本项目与 davidlinjiahao 最相似**
   - 都是 MuJoCo + ACT + SO-101
4. **视频解码是训练瓶颈**
   - torchcodec 比 pyav 快 8–20x